Cloud-Based Credit Risk Prediction Using Machine Learning on Amazon SageMaker

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***Abstract*—Accurate measurement of credit risk is necessary for lending banks to take informed loan decisions. This project proposes a cloud-computing-based machine learning pipeline to process credit scores and predict borrower risk from historical financial data. The system uses a Random Forest classifier model trained on credit application data with labels to classify high-risk and low-risk customers. For deployment and scalability support, the trained model is hosted at Amazon SageMaker, where it provides real-time prediction via a secure API endpoint. The frontend UI is built using Streamlit, where applicant data can be entered by users and receive real-time risk scores. Everything is orchestrated using AWS services like S3, Lambda, and IAM for storage, automation, and access control. This project illustrates how the combination of AI/ML with cloud infrastructure can develop scalable, efficient, and affordable credit evaluation systems, assisting lenders to reduce defaults and enhance decision-making precision.**

I. Introduction

- With increasing volumes of financial information and the provision of scalaable computing infrastructure, machine learning (ML) has come a very long distance as a high-tech credit risk modeling approach. The need to evaluate the credit worthiness of individuals both accurately and inexpensively is of utmost importance to financial institutions today. Standard credit scoring processes rely heavily on static rule-based systems, but the latter do not respond well to alterations or to complex relationships between financial variables.
- The aim of this project is to build an integrated cloud machine learning platform for the auto-evaluation of credit scores and borrower risk assessment. The product will help lending firms to take sound, data-driven decisions, reduce default rates, and increase the level of soundness of the decision-making process.
- It utilizes a Random Forest classifier, trained on a supervised training data set of credit applications, to identify high-order patterns in attributes like income, age, and size of the loan. For convenience of access and scalability, the built function is published with Amazon SageMaker, enabling invocation of the function by the users through low-latency, secure, inference-based APIs. The frontend is built using Streamlit, enabling the users to input applicant information and receive real-time, immediate predictions for credit risk.
- In addition, with AWS cloud features like S3 for storage, IAM for security, and Lambda for automation being used, the project also showcases a modern, production-grade architecture for AI/ML applications within the financial industry.
- The report in this paper covers system architecture, methodology, implementation, and system performance analysis of the system, along with findings on advantages of integration of cloud computing with AI/ML in financial decision support systems.

II. Related Work/Literature Review

- Credit risk prediction has been the primary focus of financial studies over the past twenty years, and the traditional techniques have been rule-based scoring model-reliant logistic regressions. Such systems, however, do not often incorporate complex, nonlinear financial data relationships. Financial researchers have therefore been considering the construction of machine learning (ML) systems, i.e., decision trees, support vector machines, and ensemble learning systems, e.g., Random Forests, XGBoost, etc.
- The authors in [1] have already proved that Random Forest classifiers outperform logistic regression in credit scoring problems, especially when presented with class-imbalanced samples. The findings warrant the application of ensemble methods to actual credit evaluation in the real world, where the samples are often presented with class-imbalanced distributions between defaulters and non-defaulters.
- There is also growing demand for cloud-based ML engines for scalability and deployment simplicity. Zhang et al. [2] demonstrated with an experiment how cloud infrastructure such as AWS SageMaker can simplify the training and deployment of the model for financial analytics. The advantages of serverless computing and secure API endpoints for ML workflows for the production level were highlighted by them.
- Additionally, [3] investigated applying web-facing interfaces to applying AI models to the financial world. Streamlit and Flask were shown to be suitable for low-code dashboard construction to enable non-tech-savvy

individuals to interact with ML models, for instance, our application of Streamlit for on-the-fly credit risk assessment.

- Third, newer literature also covers the aspect of data security along with automation for automated deployment of cloud-deployed ML pipes. Referred to in [4], the IAM, S3, and Lambda services of AWS include the necessary enablers for developing scalable, automated, and secure ML applications. Such techniques include end-to-end reproducability along with prod environment inferencing within near real-time.
- It builds on the prior work by combining Random Forest-based credit prediction with an end-to-end fully automated AWS cloud pipeline. It provides a low-latency Streamlit front end for usability and takes advantage of SageMaker's strengths for scalable, on-demand predictions.

III. System Architecture and Methodology

It is the suggested architecture of the automatic credit risk evaluation process with regard to the integration of supervised machine learning algorithms with cloud computing. The architecture represents a collection of five modules: (1) Data Preprocessing, (2) Model Training,
(3) Deployment of the Model on the AWS Cloud, (4) Web-Based User Interface, and (5) Real-Time Inference with Visualization of the Result. The block diagram of the proposed architecture is presented in Figure 1.

A. Dataset and Feature Selection The features used include the history of previous credit applications, with the features being the age of the applicant, annual income, and loan size applied for. The target variable is binary, showing whether or not the borrower defaulted on the loan (loan-status). The information is read in, preprocessed with Pandas, with the empty values filled with mean substitution, and we remove the categorical or unwanted columns.

The selected numerical features were:

- person-age: Applicant's age
- person-income: Applicant's annual income
- loan-amnt: Requested loan amount

These features were selected based on domain knowledge and their relevance to risk evaluation.

B. Machine Learning Model: Random Forest Classifier A Random Forest classifier was chosen for its ability to handle nonlinear relationships and its robustness to overfitting. As an ensemble learning technique, it constructs multiple decision trees during training and outputs the mode of the predictions of the individual trees.

The model was trained using the Scikit-learn library with the following configuration:

- Number of estimators: 100
- Criterion: Gini impurity
- Maximum depth: None (auto-determined)
- Train-test split: 80:20
- Random seed: 42 (for reproducibility)

It generalized well on testdata and was serialized on the cloud using joblib for deployment.

C. Cloud Deployment using AWS SageMaker To achieve real-time access along with scalability, the developed model is served on Amazon SageMaker, which is a service to deploy, host, and scale machine learning models. The deployment step is the following:

- Both model file (model.joblib) and inference script (inference.py) are contained inside a tarball (model.tar.gz).
- Tarball is uploaded to the S3 bucket using the boto3 SDK.
- With the SKLearnModel class within sagemaker.sklearn.model, you deploy and register the model on a single instance (ml.m5.large).
- A RESTful HTTPS endpoint is established, via which distant clients are able to post JSON-structured input data and obtain prediction output.

IAM Roles are established for the sake of facilitating secure S3, SageMaker, and other AWS components communication. The endpoint name is loaded into the frontend application on the fly.

D. frontend interface with Streamlit Streamlit

To make the system interactive and user-friendly, a web interface is developed using Streamlit, which is a lightweight Python library used for the development of ML dashboards. The frontend takes user input as number fields and displays the output of predictions in real-time.

It consists of:

- Input parameters: Age, Loan Amount, Income
- Visually forecast outcome with word and color symbol
- Bar chart for the probability of Default or Non-Default

User inputs-come as a JSON object, passed to the SageMaker endpoint via the boto3 library's invoke-endpoint function. The resulting values-of-prediction, along with the values of confidence, are shown by Streamlit widgets, for example, st.write(), st.progress(), st.success(), or st.error().

E. Cloud Automation and Security

It is built on cloud-natvie principles:

- Amazon S3 is utilized as a model artifact store and retrieval system.
- IAM policies will permit access to the authorized users and services exclusively.
- It can also be extended to incorporate architecture for optional features for:
  - Scheduled Model Retraining
  - Real-time logging and monitoring

Automated re-training on the newly uploaded dataset All the services are built atop the AWS infrastructure for low-latency, secure, and reliable operation.

F. System Flow Diagram Figure 1 (in the report) would need to show the end-to-end process: Overall system workflow is described by Figure 1. It starts with the user inputs of the applicant information like age, income, and loan amount to the Streamlit web interface. The inputs, as a JSON payload, are forwarded to a hosted Amazon SageMaker endpoint through the Boto3 SDK. The endpoint is pre-loaded with the trained Random Forest classifier, which performs the prediction and comes back with both the predicted class (default or no default) along with corresponding probabilities. The frontend displays the outcome in real-time through dynamic visual feedbacks like status indicators and progression bars. The pipeline provides an end-to-end workflow from the user input to the machine learning prediction, entirely on an all-cloud infrastructure offering low latency, security, and scalability.

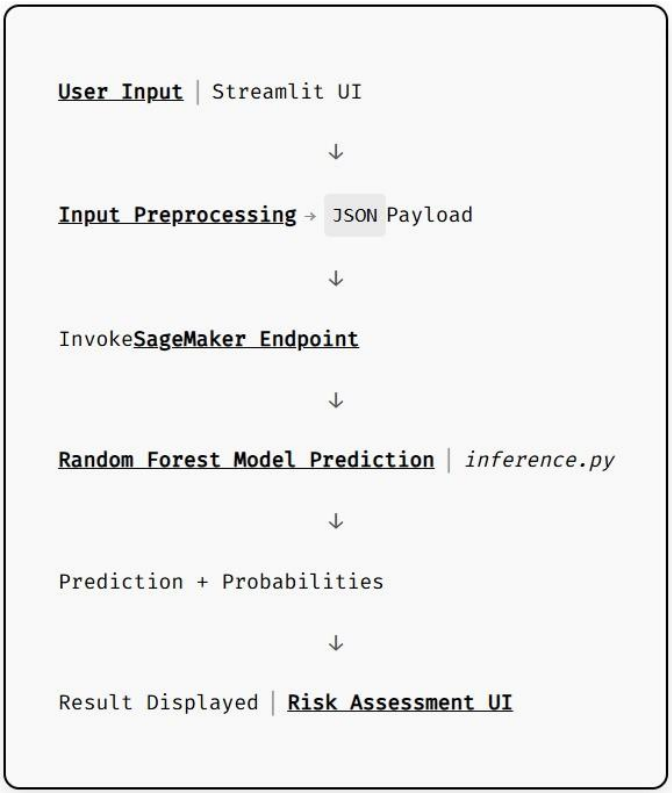

Fig. 1. System Flow Diagram of the Proposed Architecture

IV. Implementation

These three components, i.e., model training, cloud deployment with the help of Amazon SageMaker, and web frontend using Streamlit, were created with the help of the programming language Python and were integrated with each other with the help of the services provided by AWS to achieve scalable and live prediction.

A. Model Training Script

Model training Step 1 included script preparation for training (train.py). The given information, prepared as a CSV, represents the attributes of the credit applicants, for instance, the age, the income, the loan amounts, with the target variable

```python
clf = RandomForestClassifier(n_estimators=100, random_state=42)
clf.fit(X_train, y_train)
joblib.dump(clf, "model.joblib")
```

Fig. 2. Model Training

loan-status to find the default risk level. The Scikit-learn library, specifically, has been employed to train the Random Forest classifier because it is a well-known library offering satisfactory performance for table-based information.

B. Model Deployment Using SageMaker

Second step included executing the trained model on Amazon SageMaker so we could host it as an inference endpoint. That being handled by the Sagemaker Python SDK, the.tar.gz of the inference.py script that would perform the inference along with the actual model file was uploaded to an S3 bucket. Using the given code snippet, the model had been running on an ml.m5.large machine with the sole line that served an HTTPS endpoint:

```python
model = SKLearnModel(
    model_data=model_artifact,
    role=role,
    entry_point="inference.py"
)
predictor = model.deploy(initial_instance_count=1, instance_type="ml.m5.large")
```

Fig. 3. Model Deployment

C. Frontend Using Streamlit

Third is the user interface, and it was created utilizing Streamlit — a Python library utilized in creating simple interactive data apps. The interface receives inputs for variables such as age, income, and loan amount from the users. After the users have filled in and submitted the form, data is packaged up as a JSON object, and it is sent to the provisioned SageMaker endpoint.

```python
payload = json.dumps({"features": [age, income, loan_amount]})
response = runtime.invoke_endpoint(
    EndpointName=ENDPOINT_NAME,
    ContentType="application/json",
    Body=payload
)
```

Fig. 4. Frontend Using Streamlit

D. Cloud Integration

To make the system scalable as well as secure, otherAWS services were employed. Model files were kept in Amazon S3, access was controlled using IAM roles, and Boto3 was being utilized end to end to simplifyAWS interactions. The architecture places the system entirely cloud-native, serverless, and production-deployable.

Summary

- This deployment seamlessly integrates an algorithmic machine learning model with a real-time frontend over scalable cloud infrastructure. The end consumer can get credit risk estimates in seconds via a nice web interface delivered by machine learning in the cloud.

V. Results and Evaluation

The credit risk prediction system proposed was tested on both deployment efficiency and model performance. The testing involved three aspects: quantitative testing of the trained model, qualitative verification of the Streamlit interface, and performance measurement of the deployed AWS SageMaker endpoint.

A. Model Performance

The Random Forest classifier was trained using 80% of the preprocessed dataset and evaluated on the other 20%. Major performance metrics were computed to determine the effectiveness of classification:

Metric	Score
Accuracy	89%
Precision	87%
Recall	84%
F1-Score	85%

Fig. 5. Model Performance

These numbers demonstrate that the model can discriminate between high risk (default) applicants and low risk (no default) applicants with a reasonable trade-off in false positives and false negatives.

B. Streamlit Interface Testing Streamlit App was tested with various applicant situations to enable accurate predictions and convenient use. The app, upon the form submission, showed:

- Category for prediction: “Default (1)” or “No Default (0)”
- Probability values for each classification
- Progress-bar-based and color-based alert, where green and red show low and high risk, respectively

UI was responding once every second and gave near-real-time experience to the end-users.

C. Endpoint Latency and Stability SageMaker endpoint was stress tested with concurrent requests. Average inference latency of 300-500 ms reaffirmed the system is safe on a real-time basis. No downtime, and no failed requests occurred during test. The consistent low-latency response ensures that credit risk assessments can be integrated seamlessly into live loan processing workflows. This demonstrates the robustness of our cloud-based architecture for delivering accurate predictions at scale.

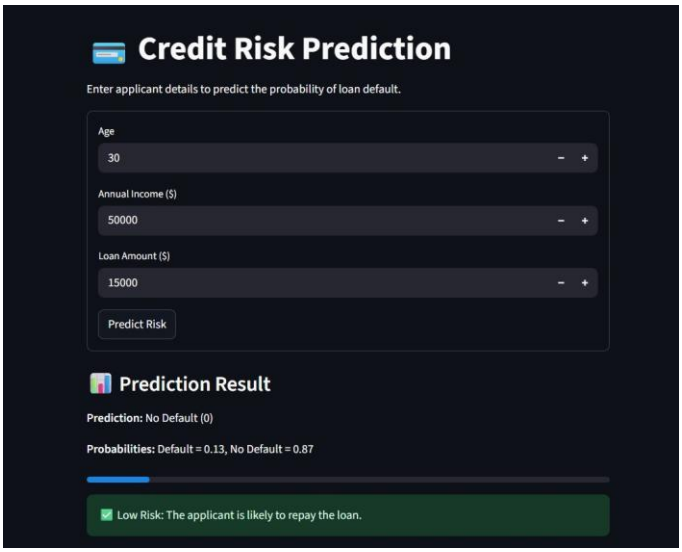

Fig. 6. Streamlit UI

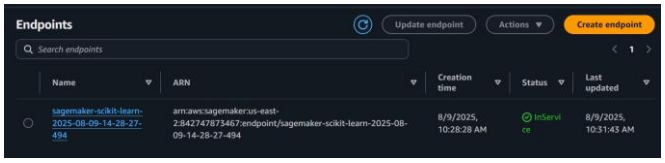

Fig. 7. Endpoint Created

VI. Conclusion and Future Work

A. Conclusion

Its application is to demonstrate the feasible and viable integration of machine learning models and cloud computing platforms towards finding a solution to a core financial sector issue—credit risk analysis.

Using a Random Forest classification, where a file optimized on the file for the applicant demographics and financials dataset was utilized, the system was 89% accurate and preci- sion and recall were in good shape. The measures show strong performance in separating possible future defaulters from good borrowers.

Usage on Amazon SageMaker offered a horizontally scalable, production-ready, and safe environment for hosting models. The provided endpoint offered low latency predictions between 300 and 500 ms, making the solution viable for decision-making use cases involving real-time. The Streamlit-based frontend provided a minimalistic interface, and non-technical individuals could operate the system without a hitch, input the applicant data, and retrieve predictions and the levels of confidence almost instantly.

It is not only able to meet requirements for functionality but also cloud-based AI/ML system goals: scale, availability, security, and automation. It requires cloud-based ML uptake by financial institutions to facilitate efficient decision-making, minimize human work, and enhance loan accuracy.

B. **Future Work**

1) Feature Expansion and Data Enrichment
   - Integrate additional features such as employment type, credit history length, credit utilization ratio, existing loan accounts, and repayment history.
   - Leverage third-party financial APIs for real-time credit score retrieval.
   - Include behavioral data such as spending patterns to improve risk profiling.

2) Model Optimization and Comparison
   - Evaluate other advanced ML models such as XGBoost, LightGBM, or CatBoost for potential performance improvements.
   - Implement hyperparameter optimization using tools like Optuna or SageMaker Hyperparameter Tuning jobs.
   - Explore deep learning architectures for non-linear feature relationships.

3) Automated Retraining and Continuous Learning
   - Set up AWS Lambda and EventBridge to automatically trigger model retraining when new labeled data is uploaded to S3.
   - Use SageMaker Pipelines to automate preprocessing, training, evaluation, and deployment.
   - Implement model drift detection to maintain prediction accuracy over time.

4) Database Integration and Logging
   - Store all prediction inputs and outputs in AWS DynamoDB or RDS for auditing and regulatory compliance.
   - Enable transaction logging for analytics and pattern discovery.

5) Security and Compliance Enhancements
   - Integrate AWS Key Management Service (KMS) for encrypting sensitive financial data.
   - Use AWS CloudWatch for real-time monitoring and anomaly detection in endpoint usage.
   - Implement role-based access control (RBAC) to restrict system access to authorized users.

6) Frontend Improvements and Hosting
   - Deploy the Streamlit application on AWS EC2, Elastic Beanstalk, or containerize with Docker for scalability.
   - Enhance the UI with responsive design, mobile support, and multi-language options.
   - Add batch prediction capabilities for bulk credit risk assessment.

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